

Fundamentals of Photovoltaics (EE336)

Lecture 3: Semiconductor and P-N junction

Semiconductor Materials

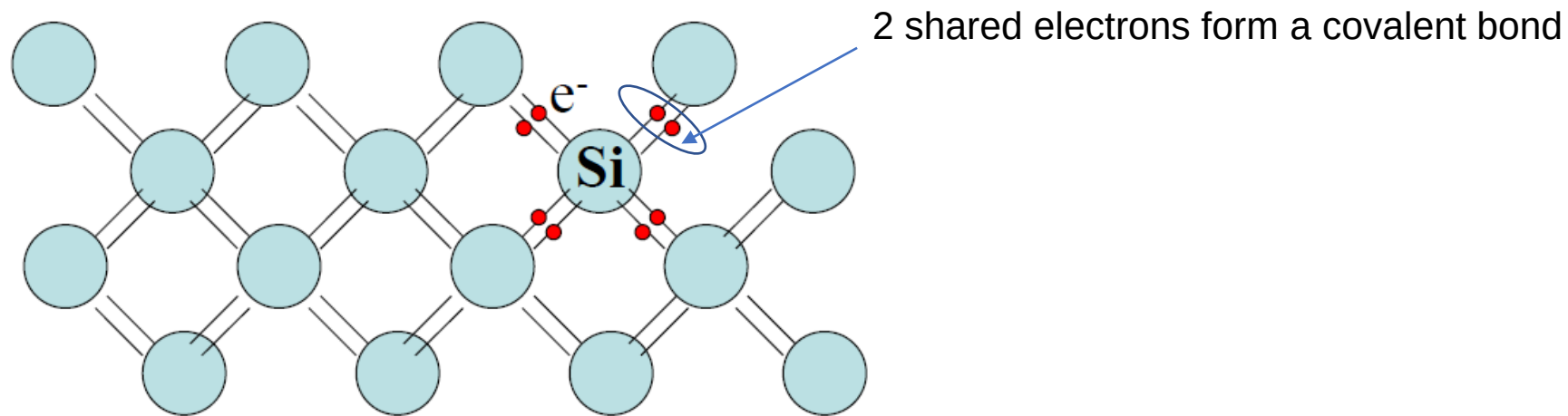
- Semiconductor materials come from different groups in the periodic table, yet share certain similarities.
- The properties of the semiconductor material are related to their atomic characteristics, and change from group to group.
- Researchers and designers take advantage of these differences to improve design and choose the optimal material for a PV application.
- The atoms in a semiconductor are materials from either group IV of the periodic table, or from a combination of group III and group V (called III-V semiconductors), or of combinations from group II and group VI (called II-VI semiconductors). Because different semiconductors are made up of elements from different groups in the periodic table, properties vary between semiconductors. Silicon, which is a group IV, is the most commonly used semiconductor material as it forms the basis for integrated circuit (IC) chips and is the most mature technology and most solar cells are also silicon based.

							VIIIA		
							² He 4.003		
		IIIA	IVA	VA	VIA	VIIA			
		⁵ B 10.811	⁶ C 12.011	⁷ N 14.007	⁸ O 15.999	⁹ F 18.998	¹⁰ Ne 20.183		
		¹³ Al 26.982	¹⁴ Si 28.086	¹⁵ P 30.974	¹⁶ S 32.064	¹⁷ Cl 35.453	¹⁸ Ar 39.948		
IB	IIB	²⁹ Cu 63.54	³⁰ Zn 65.37	³¹ Ga 69.72	³² Ge 72.59	³³ As 74.922	³⁴ Se 78.96	³⁵ Br 79.909	³⁶ Kr 83.80
		⁴⁷ Ag 107.870	⁴⁸ Cd 112.40	⁴⁹ In 114.82	⁵⁰ Sn 118.69	⁵¹ Sb 121.75	⁵² Te 127.60	⁵³ I 126.904	⁵⁴ Xe 131.30
		⁷⁹ Au 196.967	⁸⁰ Hg 200.59	⁸¹ Tl 204.37	⁸² Pb 207.19	⁸³ Bi 208.980	⁸⁴ Po (210)	⁸⁵ At (210)	⁸⁶ Rn (222)

Section from the periodic table. More common semiconductor materials are shown in blue. A semiconductor can be either of a **single element**, such as Si or Ge, a **compound**, such as GaAs, InP or CdTe, or an **alloy**, such as $\text{Si}_x\text{Ge}_{(1-x)}$ or $\text{Al}_x\text{Ga}_{(1-x)}\text{As}$, where x is the fraction of the particular element and ranges from 0 to 1.

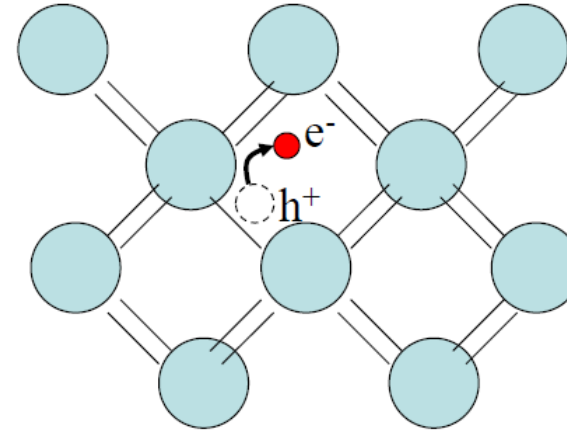
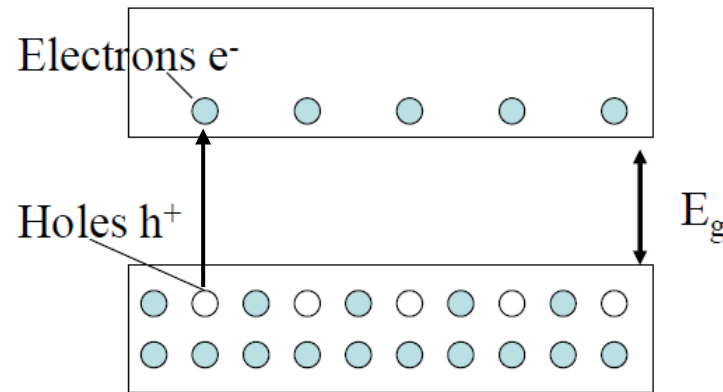
Semiconductor Structure

- Semiconductors are made up of atoms bonded together to form a uniform structure
- For example, silicon atoms has four valance electrons, which are shared with four neighbouring silicon atoms, forming covalent bonds.



Conduction in Semiconductors

- The electrons in the covalent bond formed between each atom of the atoms in the lattice structure are held in place by this bond and hence they are localized to the region surrounding the atom. These bonded electrons cannot move or change energy, thus are not considered “free” and cannot participate in current flow, absorption, or other physical processes of the interest in solar cell.
- However, the electron in this “stuck” bonded arrangement is only at absolute zero temperature.
- At elevated temperature, especially at the temperatures where solar cells operate, electrons gain enough energy to break the covalent bond and escape from the bonds. In other word, the electrons in the valence band are excited to the conduction band across the bandgap.
- Therefore, some of the states in the conduction band now occupied by electrons and some of the states in the valence band is now empty. An empty state in the valence band is referred to as a hole. It corresponds to the broken bond in the covalent bond.
- The electrons and holes can move freely (at least for a short period of time) before recombination.



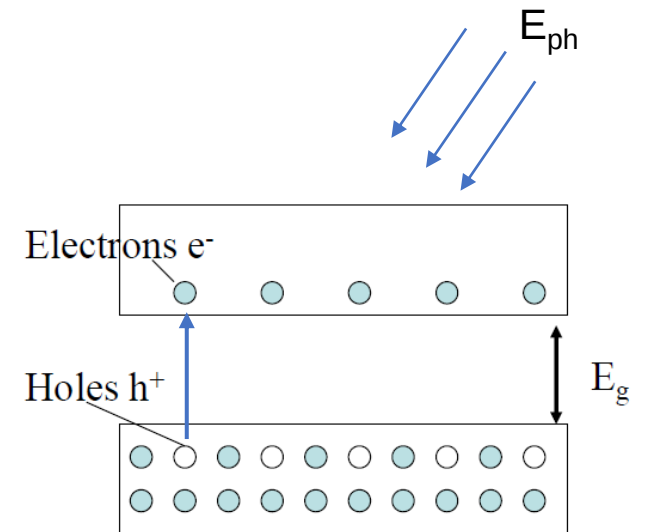
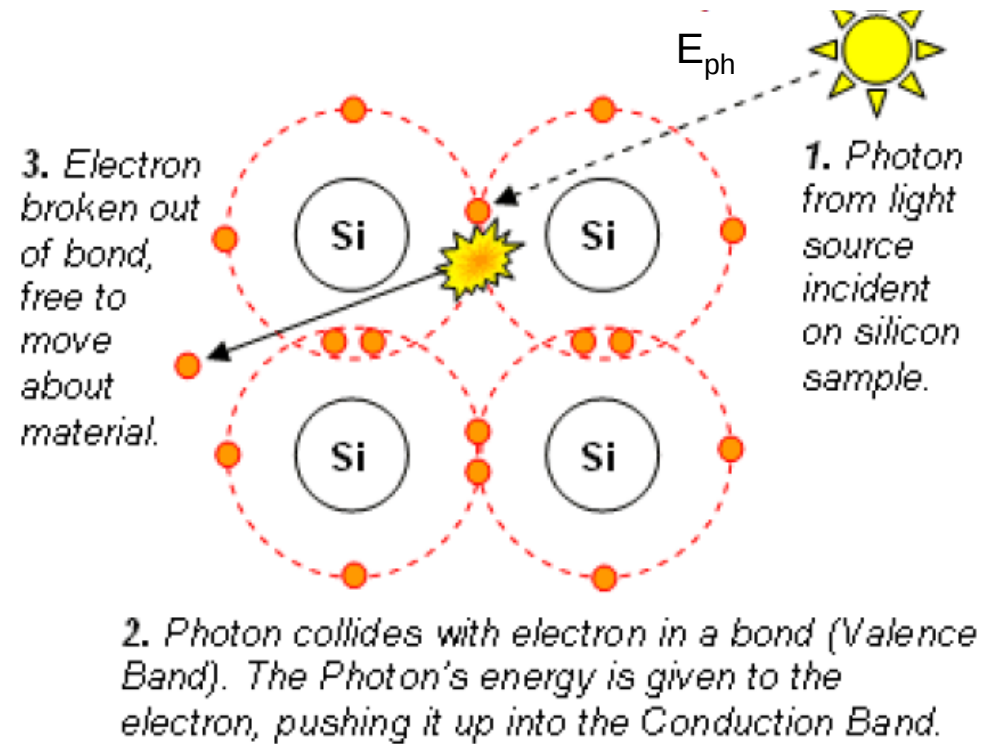
- The electrons in the valence band can also be excited to the conduction band, creating electron-hole pairs, by illuminating the semiconductor with light having a photon energy E_{ph} equal or greater than the bandgap energy E_g .

- $E_{ph} \geq E_g$

- $E_{ph} = h\nu = \frac{hc}{\lambda}$

- $\frac{hc}{\lambda} \geq E_g$

- $\frac{1240}{\lambda (nm)} \geq E_g (eV)$



Intrinsic Material

Semiconductor material which has not had impurities added to it is called intrinsic material.

The concentration of carriers (electron and holes) in intrinsic material created by the thermal excitation is called the intrinsic carrier concentration.

This number of carrier depends on the band gap of the material and on the temperature of the material.

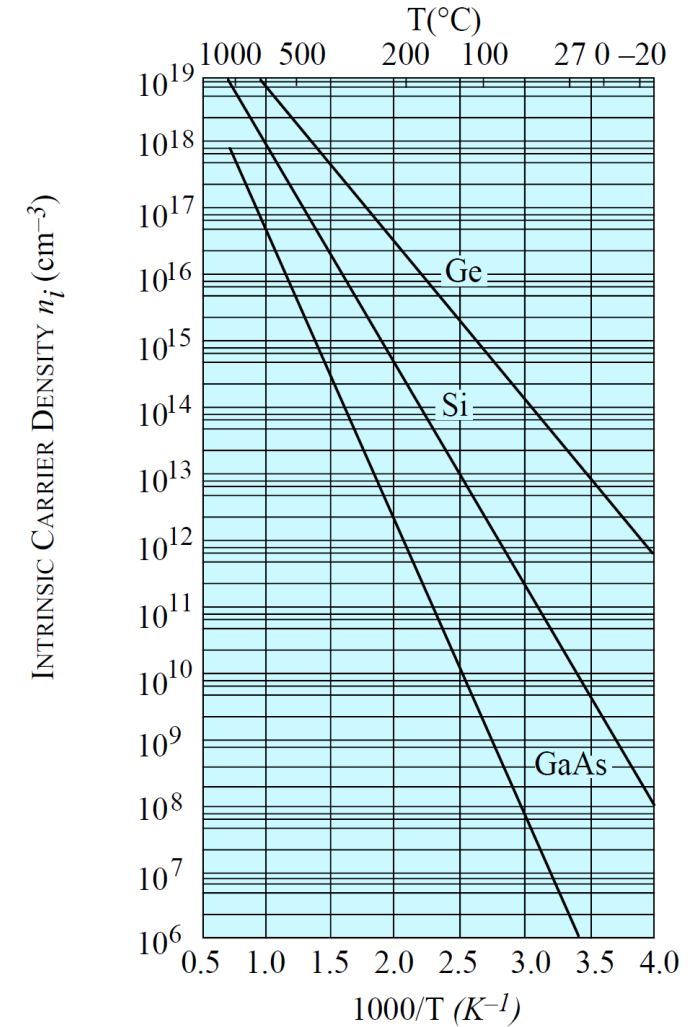
A large band gap will make it more difficult for a carrier to be thermally excited across the band gap, and therefore the intrinsic carrier concentration is lower in higher band gap materials.

Alternatively, increasing the temperature makes it more likely that an electron will be excited into the conduction band, which will increase the intrinsic carrier concentration.

$$n_i = p_i = 2 \left(\frac{k_B T}{2\pi\hbar^2} \right)^{3/2} (m_e^* m_h^*)^{3/4} \exp(-E_g/2k_B T)$$

The table lists the value of n_i for some semiconductors at 300K. We see that n_i increases with decreasing energy gap E_g . This is expected since n_i is exponentially dependent on E_g .

Materials	n_i (cm ⁻³)	E_g (eV)
GaAs	1.8×10^6	1.43
Si	1.5×10^{10}	1.11
Ge	2.4×10^{13}	0.67



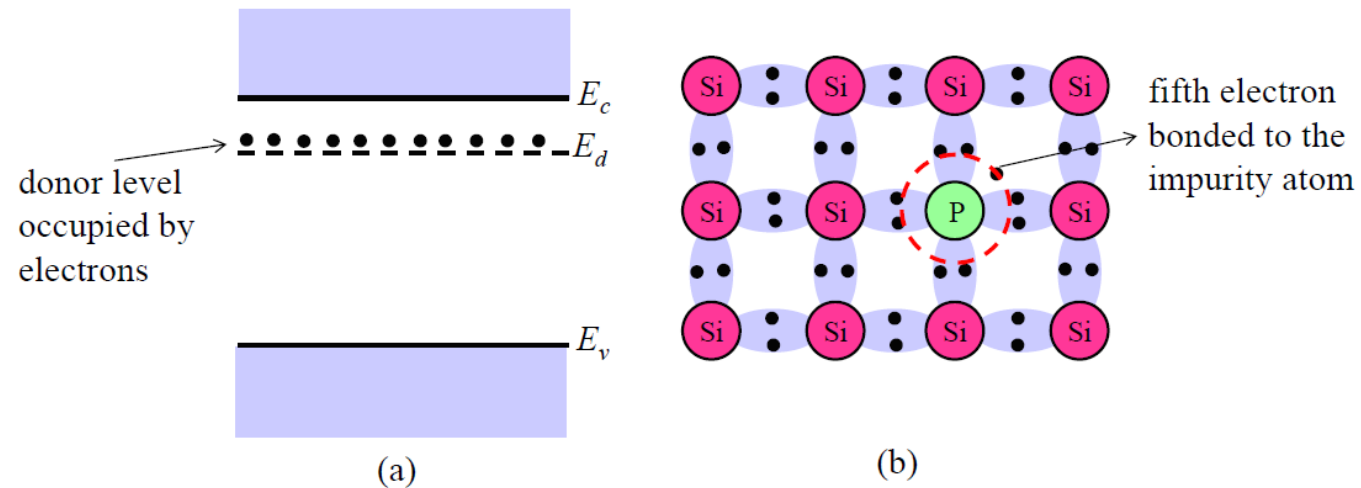
N-type Material

N-type material is created when semiconductor materials from group IV are doped with group V.

Suppose a Si is doped with phosphorous P. Si atom has 4 valence electrons, but P atom has 5 valence electrons. Four valence electrons of P atom make covalent bonds with neighbouring Si atoms. The fifth electron is held by P atom but much more weakly than covalent bonds.

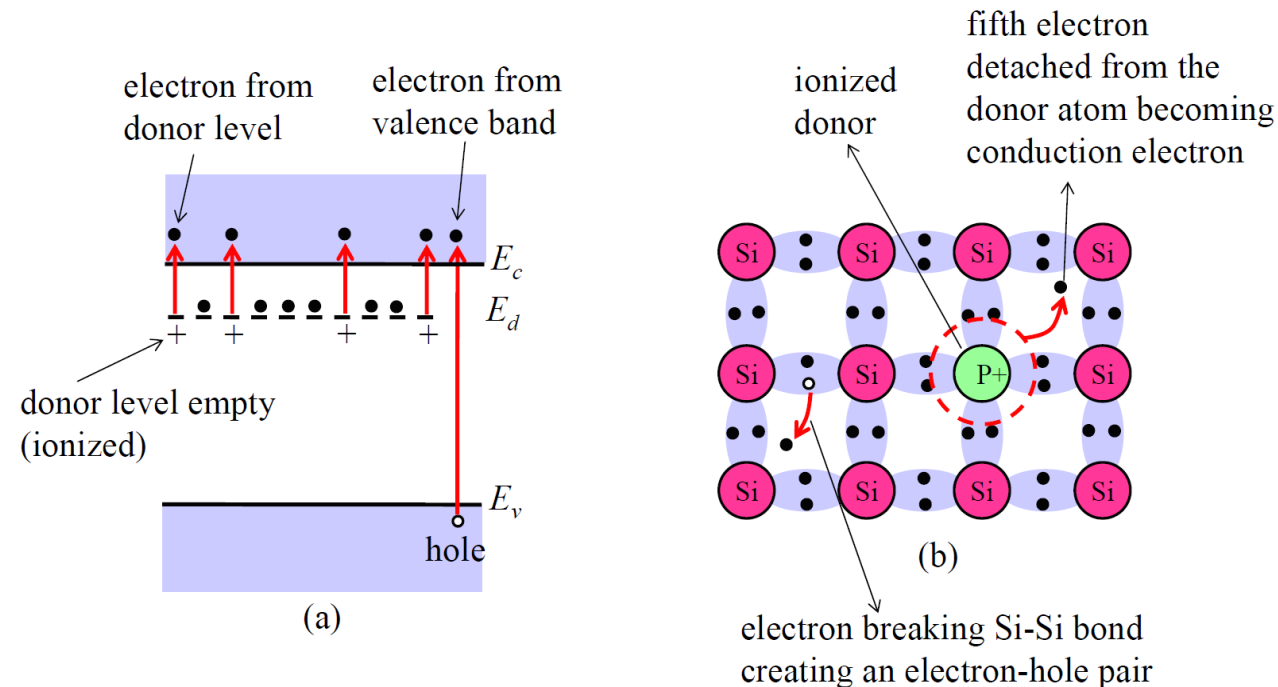
The impurity P atoms introduce an energy level called donor level (E_d) in the bandgap below the conduction band edge E_c .

At 0K, the donor level is occupied by the electrons, which corresponding to the fifth electrons being bonded to the P atom.



Above 0K, the electrons in the donor level and valence band gain thermal energy.

So, electrons from donor level are excited to the conduction band. This corresponds to the fifth electrons breaking the bonds with impurity P atoms becoming conduction electrons. The impurity atoms then become positively charged or ionized. (Note: in that case, electrons are donated to the conduction band without creating holes in the valence band.)



Relatively much smaller number of electrons from the valence band are excited to the conduction band, creating electron-hole pairs. However, the electrons excited to the conduction from the valence band is much smaller than those excited from the donor level. This is because the energy of bandgap E_g is much bigger than the energy required to excite the electrons from the donor level to the conduction band ($E_c - E_d$).

Since the excitation of electrons from the donor level to the conduction band ionizes the impurity atoms (donor atoms), $E_c - E_d$ is called ionization energy.

At temperature T whereby thermal energy is much higher than ionization energy, all the donor atoms are ionized.

This phenomenon is impurity saturation.

$T = 300 \text{ K}$ (room temperature) corresponds to impurity saturation.

At this condition, the electron concentration in the conduction band is approximately equal to concentration of donor atoms.

$$n_0 \approx N_d$$

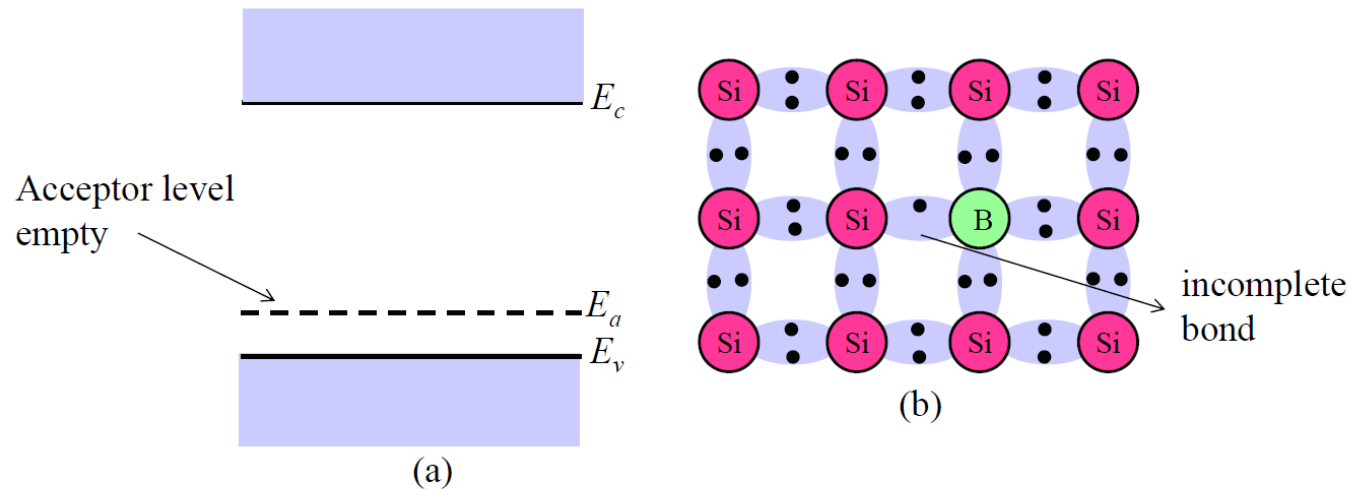
P-type Material

P-type material is created when semiconductor materials from group IV are doped with group III.

Suppose a Si is doped with boron B. Si atom has 4 valence electrons but B atom has only 3 valence electrons. In making covalent bonding with neighbouring Si atoms, the B atom only contributes 3 electrons thereby leaving one incomplete bond.

The impurity B atoms introduce an energy level called acceptor level (E_a) in the bandgap above the valence band edge E_v .

At 0K, the acceptor level is empty, which corresponding to the incomplete Si-B covalent bond.



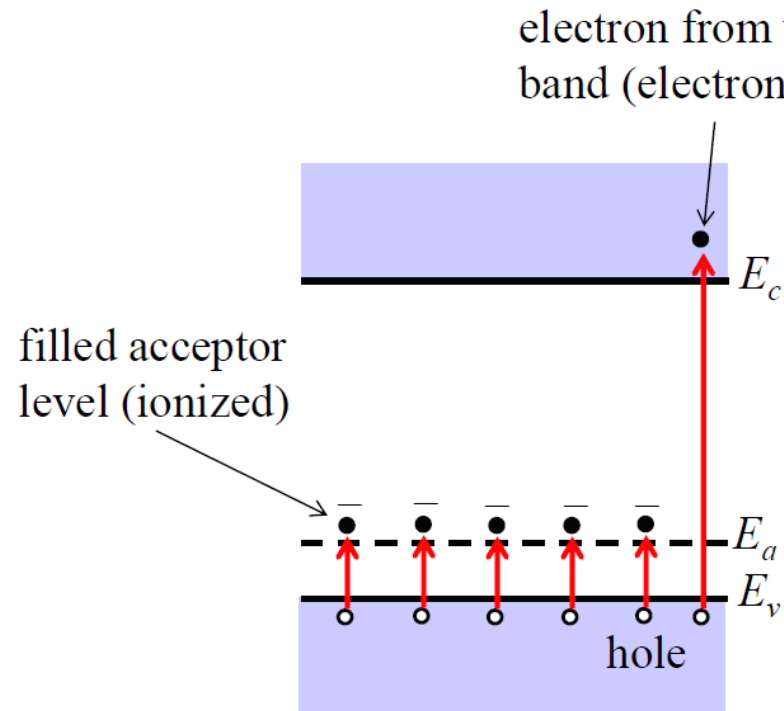
Above 0K, the electrons in the valence band gain thermal energy.

Some electrons from the valence band are excited to the acceptor level creating holes in the valence band. This corresponds to the electrons from the Si-Si bonds move occupying the incomplete Si-B bonds. The impurity B atoms then become negatively charged or ionized. (Note: in that case, holes are created in the valence band without creating electrons in the conduction band.)

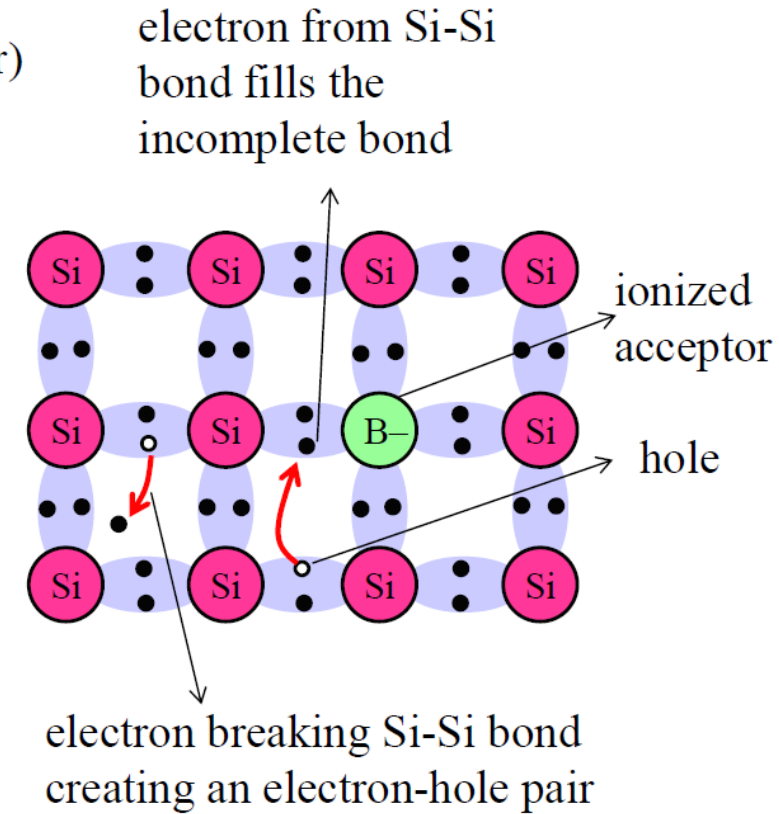
Relatively much smaller number of electrons from the valence band are excited to the conduction band, creating electron-hole pairs. This is because the energy of bandgap E_g is much bigger than the energy required to excite the electrons from the valence band to the acceptor level ($E_a - E_v$).

Herein, the ionization energy of the acceptor atoms is the energy difference $E_a - E_v$

At the thermal energy much larger than the ionization energy, the hole concentration in the valence band is approximately equal to concentration of acceptor atoms. ($p_0 \approx N_a$)



(a)



(b)

Semiconductor in Equilibrium

Equilibrium implies that no external forces, such as voltages, electric fields, magnetic fields or temperature gradients are acting on the semiconductor.

The number of carriers in the conduction and valence band at equilibrium is called the equilibrium carrier concentration. In a doped semiconductor, the equilibrium carrier concentration of the majority carrier is equal to the intrinsic carrier concentration plus the number of carriers (impurities) added. Under most conditions, the doping of the semiconductor is several orders of magnitude greater than the intrinsic carrier concentration, such that the number of majority carriers is approximately equal to the doping (accept or donor concentration).

At equilibrium, the product of the majority and minority carrier concentration is a constant, and this is mathematically expressed as

$$n_0 p_0 = n_i^2$$

where n_i is the intrinsic carrier concentration and n_0 and p_0 are the electron and hole equilibrium carrier concentrations. Using the above equation, the majority and minority carrier concentrations are given as:

$$\text{n-type, } n_0 = N_D, p_0 = \frac{n_i^2}{N_D}$$

$$\text{p-type, } p_0 = N_A, n_0 = \frac{n_i^2}{N_A}$$

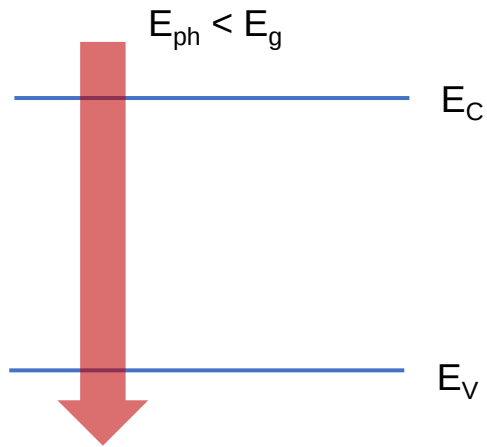
where N_D is the concentration of donor atoms and N_A is the concentration of acceptor atoms.

Light Absorption and Charge Generation

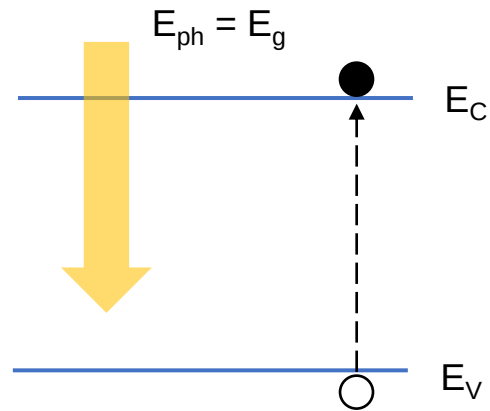
Absorption of Light

Photons incident on the surface of a semiconductor will be either reflected from the top surface, or absorbed in the material or, transmitted through the material. For photovoltaic devices, reflection and transmission are typically considered loss mechanisms as photons which are not absorbed do not generate power. If the photon is absorbed, it has the possibility of exciting an electron from the valence band to the conduction band. A key factor in determining if a photon is absorbed or transmitted is the energy of the photon. Therefore, only if the photon has enough energy will the electron be excited into the conduction band from the valence band. Photons falling onto a semiconductor material can be divided into three groups based on their energy compared to that of the semiconductor band gap:

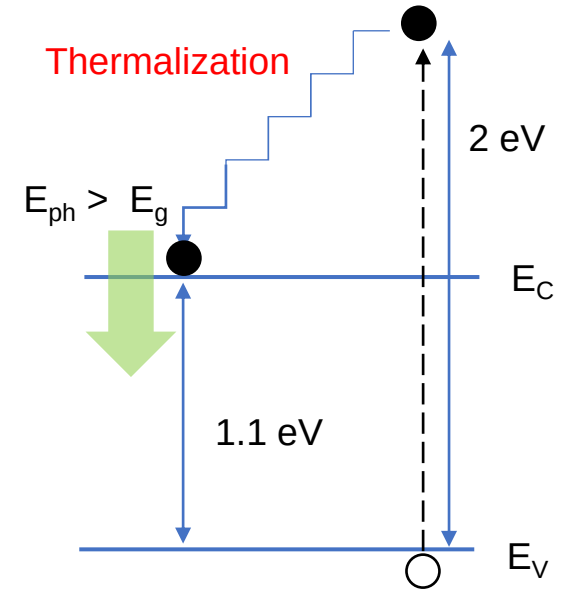
- $E_{ph} < E_g$ Photons with energy E_{ph} less than the band gap energy E_g interact only weakly with the semiconductor, passing through it as if it were transparent.
- $E_{ph} = E_g$ have just enough energy to create an electron hole pair and are efficiently absorbed.
- $E_{ph} > E_g$ Photons with energy much greater than the band gap are strongly absorbed. However, for photovoltaic applications, the photon energy greater than the band gap is wasted as electrons quickly thermalize back down to the conduction band edges.



Photon is not absorbed



No photon energy loss



Photon energy is lost in thermalization process.

Example, $E_{ph} = 2 \text{ eV}$, $E_g = 1.1 \text{ eV}$.

Energy loss = $2 \text{ eV} - 1.1 \text{ eV} = 0.9 \text{ eV}$

The absorption of photons creates both a majority and a minority carrier.

In many photovoltaic applications, the number of light-generated carriers are of orders of magnitude less than the number of majority carriers already present in the solar cell due to doping.

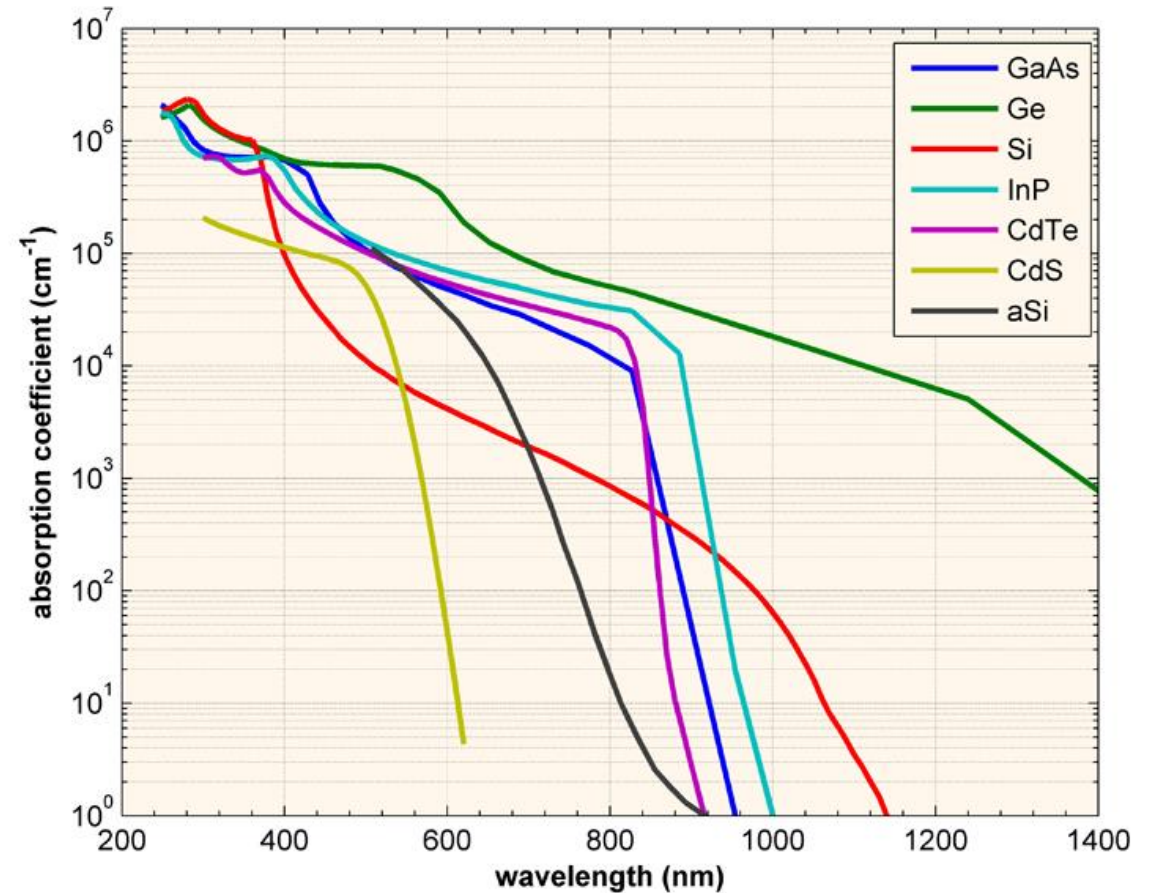
Consequently, the number of **majority carriers in an illuminated semiconductor does not alter significantly**.

However, the opposite is true for the number of minority carriers. The **number of photo-generated minority carriers outweighs the number of minority carriers** existing in the doped solar cell in the dark (because in doping the minority carrier concentration is so small), and therefore the number of minority carriers in an illuminated solar cell can be approximated by the number of light generated carriers.

Absorption Coefficient

The absorption coefficient determines how far into a material light of a particular wavelength can penetrate before it is absorbed.

In a material with a low absorption coefficient, light is only poorly absorbed, and if the material is thin enough, it will appear transparent to that wavelength. The absorption coefficient depends on the material and also on the wavelength of light which is being absorbed. Semiconductor materials have a sharp edge in their absorption coefficient, since light which has energy below the band gap does not have sufficient energy to excite an electron into the conduction band from the valence band. Consequently this light is not absorbed. The absorption coefficient for several semiconductor materials is shown in the figure.



- The graph shows that even for those photons which have an energy above the band gap, **the absorption coefficient is not constant, but still depends strongly on wavelength**. The probability of absorbing a photon depends on the likelihood of having a photon and an electron interact in such a way as to move from one energy band to another. For photons which have an energy very close to that of the band gap, the absorption is relatively low since only those electrons directly at the valence band edge can interact with the photon to cause absorption. As the photon energy increases, not just the electrons already having energy close to that of the band gap can interact with the photon. Therefore, a larger number of electrons can interact with the photon and result in the photon being absorbed.
- The absorption coefficient, α , is related to the extinction coefficient, k , by the following formula:

$$\alpha = \frac{4\pi k}{\lambda}$$

- where λ is the wavelength. If λ is in nm, multiply by 10^7 to get the absorption coefficient in the units of cm^{-1} .

How to extract absorption coefficient (α) from the absorption spectrum?

$$A = \log \left(\frac{I_0}{I} \right)$$

where, A is the absorbance which can be obtained from the absorption spectrometer.

The spectrometer use the above formula to calculate the absorbance, where is I_0 the initial light intensity of light source and I is the remaining light intensity after passing through the sample.

According to Lambert-Beer Law

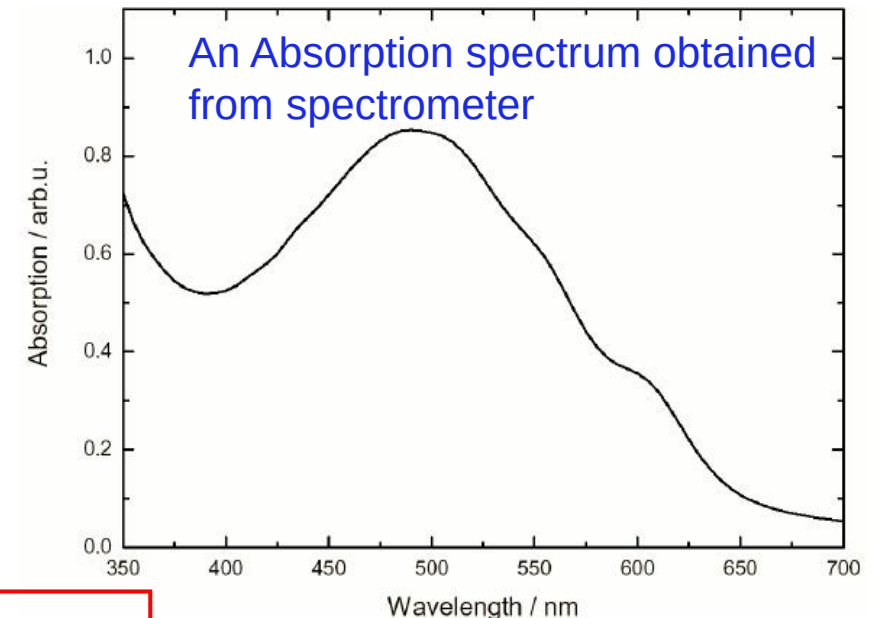
$$\alpha x = \ln \left(\frac{I_0}{I} \right)$$

Where x is the thickness of the sample.

$$\log_a b = \frac{1}{\log_b a}$$
$$\log_c a = \frac{\log_b a}{\log_b c}$$

From above two equations, $\alpha x = A * \ln 10 = 2.3 * A$

Therefore, absorption coefficient = $2.3 * \text{Absorbance} / \text{thickness of the sample}$



Absorption Depth

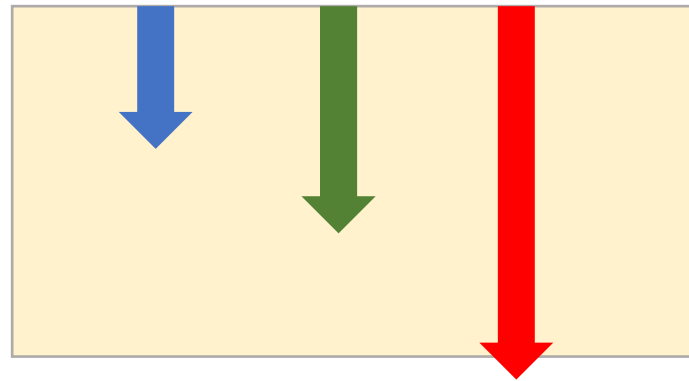
The relationship between absorption coefficient and wavelength makes it so that different wavelengths penetrate different distances into a semiconductor before most of the light is absorbed.

The absorption depth is given by the inverse of the absorption coefficient, or α^{-1} .

$$L_a = \alpha^{-1}$$

The absorption depth is a useful parameter which gives the distance into the material at which the light drops to about 36% of its original intensity, or alternately has dropped by a factor of $1/e$.

Since high energy light (short wavelength), such as blue light, has a large absorption coefficient, it is absorbed in a short distance (for silicon solar cells within a few microns) of the surface, while red light (lower energy, longer wavelength) is absorbed less strongly. Even after a few hundred microns, not all red light is absorbed in silicon.



Generation Rate

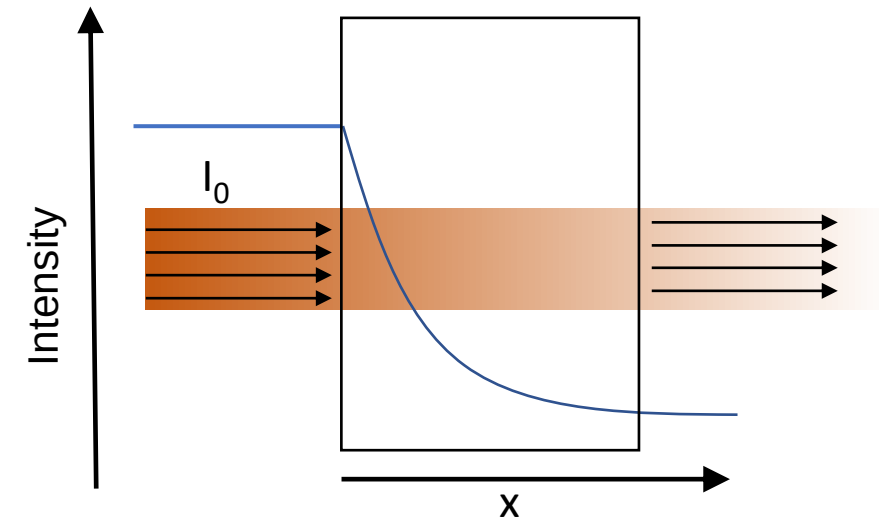
The generation rate, G (cm^{-3}) gives the number of photogenerated carriers (electron-hole pairs) generated at each point inside a bulk semiconductor due to the absorption of photons. Generation is an important parameter in solar cell operation.

Neglecting reflection, the amount of light which is absorbed by a material at a particular wavelength depends on the absorption coefficient (α in cm^{-1}) and the thickness of the absorbing material. The intensity of light at any point inside a bulk semiconductor can be calculated according to the equation:

$$I(\lambda, x) = I_0(\lambda)e^{-\alpha(\lambda)x}$$

where α is the absorption coefficient typically in cm^{-1} ;
 x is the distance into the semiconductor at which the light intensity is being calculated; and
 I_0 is the light intensity at the top surface.

The above equation is well-known as Lambert-Beer Law.



Assuming that the change in the light intensity inside the semiconductor due to the absorption of photons directly causes the generation of electron-hole pairs, then the generation rate G inside the semiconductor is determined by finding the change in light intensity along the x -distance. Consequently, differentiating the above equation will give the generation rate at any point in the semiconductor. Hence:

$$G(\lambda, x) = \alpha(\lambda) \phi_0(\lambda) e^{-\alpha(\lambda)x}$$

where ϕ_0 = photon flux at the surface (photons/unit-area/sec.);
 α = absorption coefficient; and
 x = distance into the material.

The above equations show that the light intensity exponentially decreases throughout the material and further that the generation is highest at the surface of the material.

If the reflection loss from the front surface is taken into account, the generation rate will be

$$G(\lambda, x) = [1 - R(\lambda)] \alpha(\lambda) \phi_0(\lambda) e^{-\alpha(\lambda)x}$$

where $R(\lambda)$ is the reflectance at the surface.

For photovoltaic applications, the incident light consists of a combination of many different wavelengths, and therefore the total generation rate can be calculated by integrating the entire absorption spectrum.

Recombination

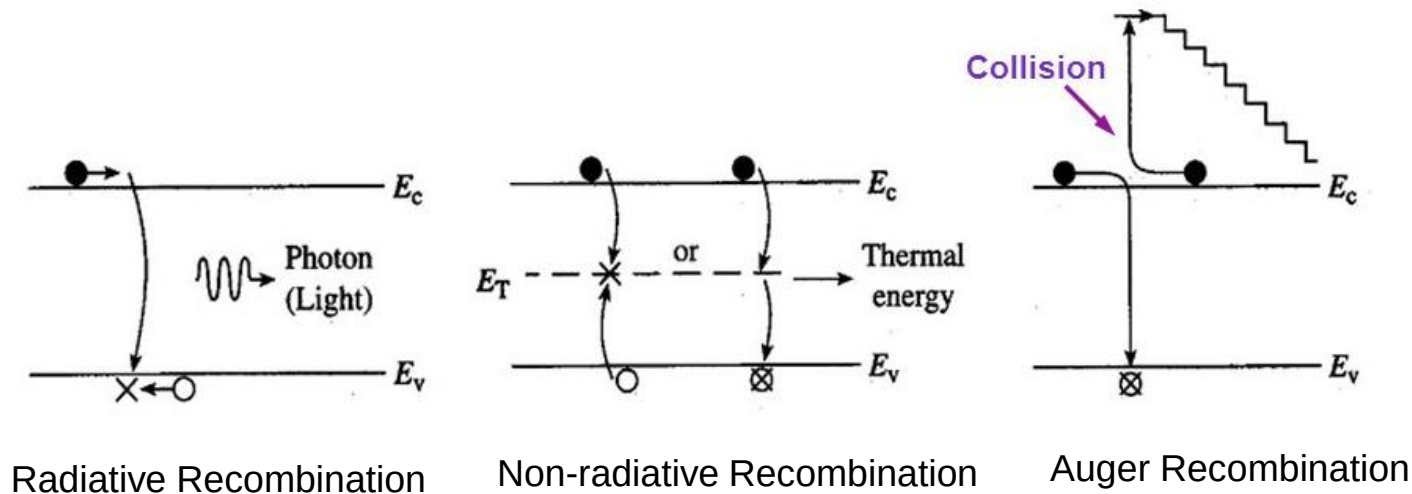
Any electron which exists in the conduction band is in a meta-stable state and will eventually stabilize to a lower energy position in the valence band. When this occurs, it must move into an empty valence band state. Therefore, when the electron stabilizes back down into the valence band, it also effectively removes a hole. This process is called recombination.

In a solar cell, recombination acts to restore the non-equilibrium photogenerated carriers to its thermal equilibrium. Three types of recombination in a bulk semiconductor are:

Radiative recombination: resulting in the emission of a photon. This is often called band-to-band recombination

Non-radiative recombination: resulting in phonon emission. This is often called trap-assisted recombination or Shockley-Read-Hall Recombination

Auger recombination resulting when kinetic energy is imparted to a third carrier. This is also a non-radiative process.

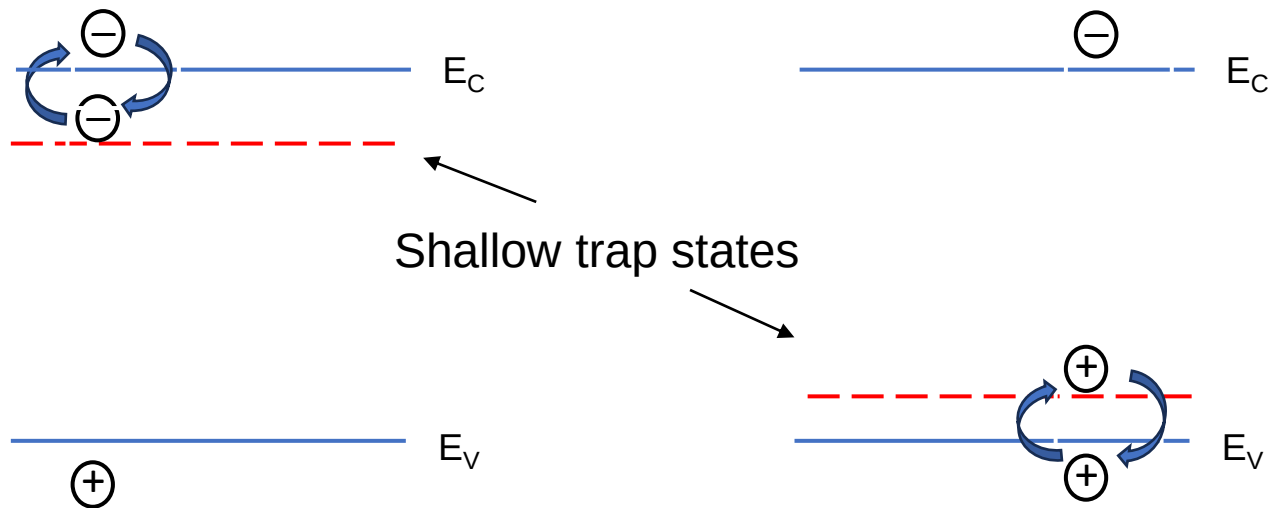


Radiative (Band-to-Band) Recombination

- Radiative recombination is the recombination mechanism that dominates in *direct* bandgap semiconductors. The light produced from a light emitting diode (LED) is the most obvious example of radiative recombination in a semiconductor device. Concentrator and space solar cells are typically made from direct bandgap materials (GaAs etc.) and radiative recombination dominates. However, most terrestrial solar cells are made from silicon, which is an *indirect* bandgap semiconductor and radiative recombination is extremely low and usually neglected. The key characteristics of radiative recombination are:
- In radiative recombination, an electron from the conduction band directly combines with a hole in the valence band and releases a photon; and
- The emitted photon has an energy similar to the band gap and is therefore only weakly absorbed such that it can exit the piece of semiconductor.

Non-Radiative Recombination

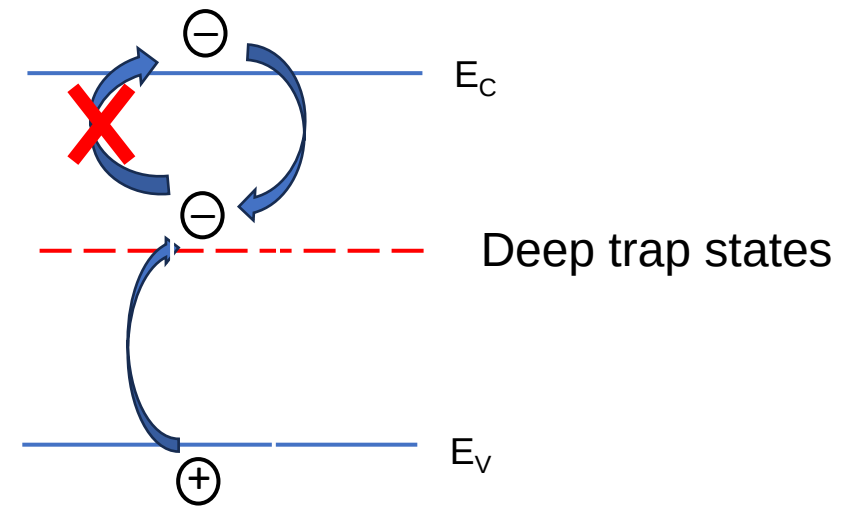
- Most practical semiconductors will have some degree of impurities or vacancies that are incorporated into the crystal structure during material growth. These impurities lead to spatially **localized defect levels** within the bandgap of the bulk semiconductor, which are often referred to as **trap states**.
- The recombination through defects/trap states is called non-radiative recombination and also known as **Shockley-Read-Hall** or **SRH recombination**. This type of recombination is often the practical limiting loss mechanism in most solar cells.
- Non-radiative recombination is a two-step process. The two steps involved are:
 1. An electron (or hole) is trapped by an energy state in the forbidden region which is introduced through defects in the crystal lattice.
 2. If a hole (or an electron) moves up to the same energy state before the electron is thermally re-emitted into the conduction band, then it recombines.
- The rate at which a carrier moves into the energy level in the forbidden gap depends on the distance of the introduced energy level from either of the band edges. Therefore, **if an energy is introduced close to either band edge, recombination is less likely** as the electron is likely to be re-emitted to the conduction band edge rather than recombine with a hole which moves into the same energy state from the valence band. For this reason, **energy levels near mid-gap are very effective for recombination**.



Shallow trap states

Defect creates energy level close to the band edge.

Non-radiative recombination is less likely to occur.



Deep trap states

Defect creates energy level near the mid-gap.

Non-radiative recombination is likely to occur.

Auger Recombination

- Auger Recombination involves three carriers. An electron and a hole recombine, but rather than emitting the energy as heat or as a photon, the energy is given to a third carrier, an electron in the conduction band. This electron then thermalizes back down to the conduction band edge.
- Auger recombination is most important at high carrier concentrations caused by heavy doping or high-level injection under concentrated sunlight. In silicon-based solar cells (the most popular), Auger recombination limits the lifetime and ultimate efficiency. The more heavily doped the material is, the shorter the Auger recombination lifetime.

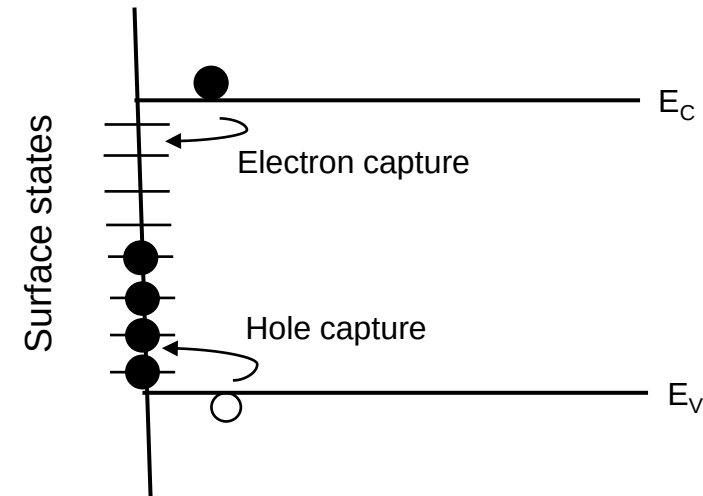
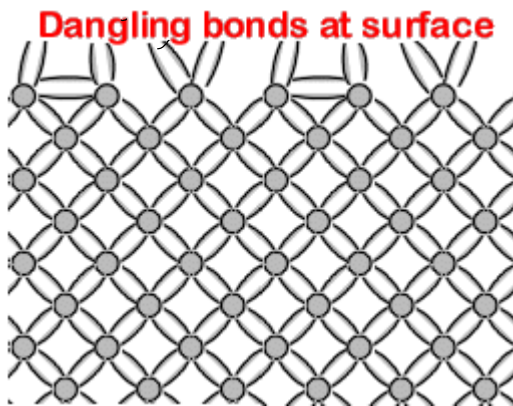
Surface Recombination

Any defects or impurities within or at the surface of the semiconductor promote recombination. Since the surface of the solar cell represents a severe disruption of the crystal lattice, the surfaces of the solar cell are a site of particularly high recombination.

The defects at a semiconductor surface are caused by the interruption to the periodicity of the crystal lattice, which causes dangling bonds at the semiconductor surface.

The dangling bonds at the surface of semiconductor cause a high local recombination rate.

The reduction of the number of dangling bonds, and hence surface recombination, is achieved by [growing a layer on top of the semiconductor surface which ties up some of these dangling bonds](#). This reduction of dangling bonds is known as [surface passivation](#).



The high recombination rate in the vicinity of a surface depletes this region of minority carriers. A localized region of low carrier concentration causes carriers to flow into this region from the surrounding, higher concentration regions.

Therefore, the surface recombination rate is limited by the rate at which minority carriers move towards the surface.

A parameter called the "surface recombination velocity", in units of cm/sec, is used to specify the recombination at a surface.

In a surface with no recombination, the movement of carriers towards the surface is zero, and hence the surface recombination velocity is zero.

In a surface with infinitely fast recombination, the movement of carriers towards this surface is limited by the maximum velocity they can attain, and for most semiconductors is on the order of 1×10^7 cm/sec.

Carrier Lifetime

- If the number of minority carriers is increased above that at equilibrium by some transient external excitation (such as incident sun), the excess minority carriers will decay back to that equilibrium carrier concentration due to and through the process of recombination. A critical parameter in a solar cell is the rate at which recombination occurs. Such a process, known as the "recombination rate" depends on the number of excess minority carriers. If for example, there are no excess minority carriers, then the recombination rate must be zero. Two parameters that are integral to recombination rate are the **minority carrier lifetime** and the **minority carrier diffusion length**.
- The minority carrier lifetime of a material, denoted by τ_n or τ_p , is the average time which a carrier can spend in an excited state after electron-hole generation before it recombines.
- Depending on the structure, solar cells made from wafers with long minority carrier lifetimes will usually be more efficient than cells made from wafers with short minority carrier lifetimes.
- The low-level injected material (where the number of minority carriers is less than the doping) the lifetime is related to the recombination rate by:

$$\tau = \frac{\Delta n}{R}$$

where τ is the minority carrier lifetime, Δn is the excess minority carriers concentration and R is the recombination rate.

Diffusion Length

- The second related parameter to recombination rate, the "minority carrier diffusion length," is the average distance a carrier can move from point of generation until it recombines. The [diffusion length](#) is closely related to [the collection probability](#).
- The minority carrier lifetime and the diffusion length depend strongly on the type and magnitude of recombination processes in the semiconductor. For many types of silicon solar cells, SRH recombination is the dominant recombination mechanism. The recombination rate will depend on the number of defects present in the material, so that as **doping the semiconductor increases the defects in the solar cell, doping will also increase the rate of SRH recombination**. In addition, since Auger recombination is more likely in heavily doped and excited material, the recombination process is itself enhanced as the doping increases. The method used to fabricate the semiconductor wafer and the processing also have a major impact on the diffusion length.
- In silicon, the lifetime can be as high as **1 msec**. For a single crystalline silicon solar cell, the diffusion length is typically **100-300 μm** . These two parameters give an indication of material quality and suitability for solar cell use.
- The diffusion length is related to the carrier lifetime by the diffusivity according to the following formula:

$$L = \sqrt{D\tau}$$

where:

L is the diffusion length in meters;

D is the diffusion coefficient in m^2/s and

τ is the lifetime in seconds.

Diode Current

We've known the information about minority carrier concentrations at both p and n-side under forward bias and low-level injection. Now we ready to formulate the diode current.

Consider the total current at a particular point (e.g. x_{n0}). It consists of two types of carriers: majority electrons ($I_n(x_{n0})$) and minority holes ($I_p(x_{n0})$).

$$I(x_{n0}) = I_n(x_{n0}) + I_p(x_{n0})$$

Let's consider minority hole current $I_p(x_{n0})$. It has diffusion and drift components. However, the drift component can be neglected because holes are minority carriers meaning that its concentration is low and the electric field is ~ 0 in the neutral region (most of the applied bias drops across W and only negligible voltage drops across the neutral p and n regions).

$$\text{Therefore, } I(x_{n0}) = I_n(x_{n0}) + I_p(x_{n0})_{\text{diff}}$$

To find the majority electron current $I_n(x_{n0})$ at x_{n0} , the edge of W on the n-side, we have to look at the minority electron current at the other edge of W , at $-x_{p0}$, on the p-side. This current is $I_n(-x_{p0})$, which is nothing but electron diffusion current at $-x_{p0}$.

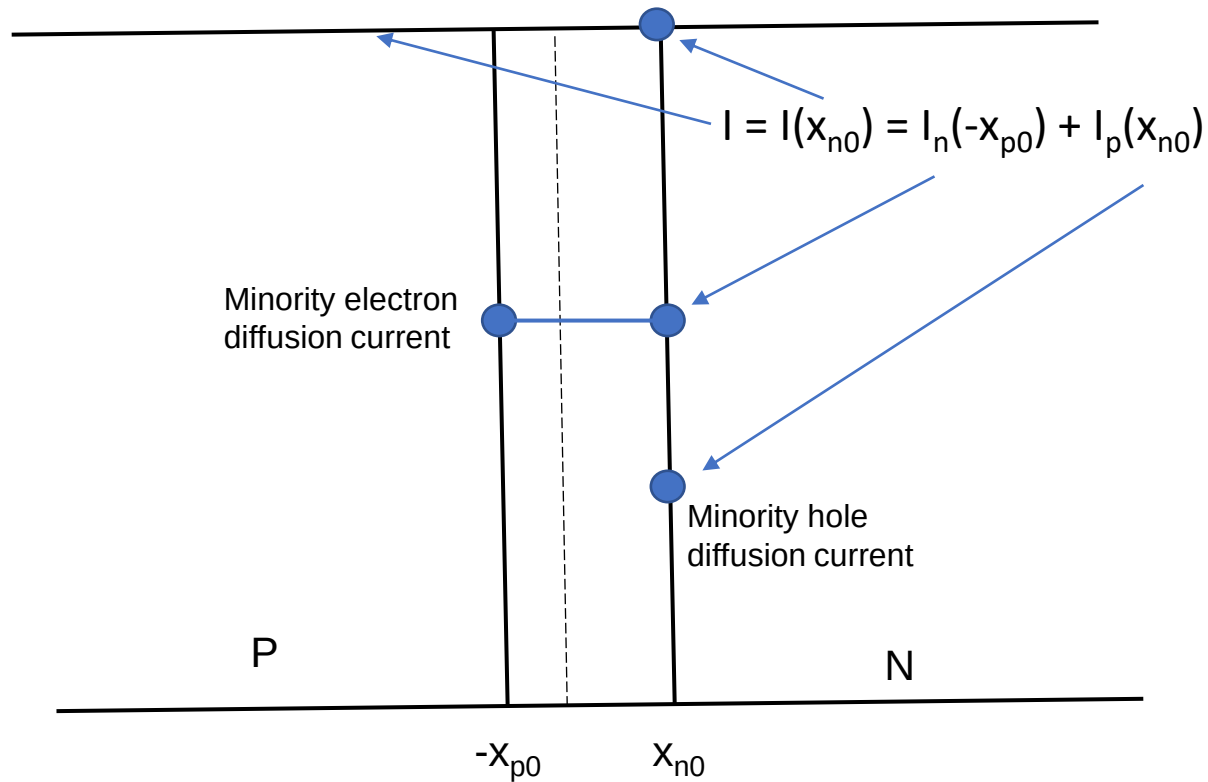
Recall that the depletion region W extends from $-x_{p0}$ to x_{n0} and in the depletion region, there is no recombination or generation of electrons. This means that the minority electron current $I_n(-x_{p0})$ must be equal to the majority electron current $I_n(x_{n0})$.

$I_n(-x_{p0})$ is a minority electron current with both diffusion and drift components but we can neglect its drift component simply because the electric field is ~ 0 out the W .

Thus, the total current I at $x = x_{n0}$ is

$$\begin{aligned} I(x_{n0}) &= I_n(x_{n0}) + I_p(x_{n0}) \\ &= I_n(-x_{p0}) + I_p(x_{n0}) \\ &\approx I_n(-x_{p0})_{\text{diff}} + I_p(x_{n0})_{\text{diff}} \end{aligned}$$

This current is, simply, a sum of the two minority carrier diffusion currents at the edges of depletion region W .



The minority diffusion currents at $-x_{p0}$ and x_{n0} can be calculated using diffusion current equation ($I_n = qAD_n \frac{dn}{dx}$, $I_p = -qAD_p \frac{dp}{dx}$) and setting $x = 0$.

$$\begin{aligned} I(x_{n0}) &= \frac{qAD_n}{L_n} \Delta n_p + \frac{qAD_p}{L_p} \Delta p_n \\ &= qA \left[\frac{D_n}{L_n} n_{p0} + \frac{D_p}{L_p} p_{n0} \right] \left[e^{\frac{qV}{kT}} - 1 \right] = I_0 \left[e^{\frac{qV}{kT}} - 1 \right] \end{aligned}$$

where I_0 is called **the reverse saturation** or generation current.

$$I_0 = qA \left[\frac{D_n}{L_n} n_{p0} + \frac{D_p}{L_p} p_{n0} \right]$$

Since the current flow must be the same at every point in the p-n junction, the total current flowing at $x = x_{n0}$ ($I(x_{n0})$) must also be the current flowing through all x . The diode current equation

$$I = I_0 \left[e^{\frac{qV}{kT}} - 1 \right]$$

is called the ideal diode equation that depicts the current of the p-n junction as a function of the bias voltage V . The equation is valid for both FB and RB.

The above equation is for ideal diode. In reality, the diode current can be deviated from the ideal diode current because of the recombination in the bulk and surface of semiconductor. The equation for non-ideal case is

$$I = I_0 \left(e^{\frac{qV}{nkT}} - 1 \right)$$

where n is the ideality factor, a number between 1 and 2 which typically increases as the current decreases.

Diode Equation for Photovoltaics

- The diode equation previously discussed is only for the current in the dark. If we want to calculate current in solar cell under illumination, photo-generation and recombination have to be taken into account.
- The derivation of diode equation for photovoltaics will not be discussed in the lecture but details are shown in Appendix (Ideal Diode Equation, total 22 page).
- The concept is similar to diode current in the dark, except that continuity equation is included for generation and recombination of photo-generated carriers. Basically, the diode equation is solved using Poisson's equation, transport equations and continuity equations with some boundary conditions.
- For the wide base p-n junction where the solar cell surface is far enough away from the junction edges that recombination properties of carriers injected into the neutral region under forward bias are not impacted, the total current is

$$\begin{aligned} I_{total} &= \left(q \frac{D_n}{L_n} n_{p0} + q \frac{D_p}{L_p} p_{n0} \right) \left(e^{\frac{qV}{kT}} - 1 \right) - qG(L_n + L_p + W) \\ &= \left(q \frac{D_n n_i^2}{L_n N_A} + q \frac{D_p n_i^2}{L_p N_D} \right) \left(e^{\frac{qV}{kT}} - 1 \right) - qG(L_n \tau_n + L_p \tau_p) \end{aligned}$$

Typically, we write the equation in the form:

$$I = J_0 \left(e^{\frac{qV}{kT}} - 1 \right) - J_{sc} \quad \text{or} \quad I = I_0 \left(e^{\frac{qV}{kT}} - 1 \right) - I_{sc}$$

Where, $J_0 = q \left[\frac{D_n}{L_n} n_{p0} + \frac{D_p}{L_p} p_{n0} \right]$

The current equation for solar cell is just a diode equation in which **photogenerated current J_{sc} is superimposed on the diode (dark) current**. J_{sc} depends on **the generation rate G** and **lifetime of the carriers (τ_p/τ_n)**.